BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE – PILANI, HYDERABAD CAMPUS

EEE/INSTR F342: POWER ELECTRONICS, SEM-II, 2026

TUTORIAL- 9 Date: 25-03-2026

Inverters

Q1. A single-phase half-bridge inverter has a resistive load of R=2.4Ω and the DC input voltage of 48V. Determine (i) the RMS output voltage at the fundamental frequency (ii) the power dissipated as heat (iii) the average and peak currents of each transistor (iv) peak reverse blocking voltage of each transistor (v) the THD of the output voltage.

Q2. For a half bridge single-phase inverter with a switching sequence of a square wave with a frequency of 60Hz, develop an expression for the load current if the load connected is an RL load of R = 10Ω and L = 25mH with a DC input voltage of 48 V. Determine the minimum and maximum inductor current.

Q3. A single-phase half-bridge inverter has an RL load of R=10Ω, L = 25mH and a DC input of 100 V. It generates a square wave of frequency 60Hz. Determine the power absorbed by the load. Calculate the THD of the load current and load voltage.

Q4. A single-phase full-bridge inverter has a resistive load of R=3Ω and the DC input voltage of 30V. Determine (i) the RMS output voltage at the fundamental frequency (ii) the power dissipated as heat (iii) the average, RMS and peak currents of each transistor (iv) peak reverse blocking voltage of each transistor (v) the % THD of the output voltage.